Phase 2 ELM Work Report

Employee Leave Management module · Fusion-Integrated

Overview. Phase 2 expanded and corrected the Employee Leave Management (ELM) module: implementing domain rules and request workflows, connecting backend and frontend surfaces, integrating permissions and routes, and adding the initial database migration. The work builds on a Phase-1 scaffold.

Scope and repository evidence

The Phase-2 commit is 04a7c6648fc487f47ddf3548421dd72f2a7c7395 (“feat(leave): Phase 2 — fully implemented and corrected ELM module”). Compared with the repository's main branch, the Phase-1 and Phase-2 commits together record 46 changed files, approximately 4,815 insertions and 154 deletions. This is an aggregate repository diff, not a count of independent issues or hours of work.

Work completed

Application functionality and user interface

- Added leave application, personal leave records, approval inbox, administration, and substitute-request pages.
- Added leave status and print components, frontend API integration, TypeScript types, and module routes/registration.

Backend, domain rules, and workflows

- Implemented API serializers, views, and URL routes for leave operations.
- Added leave-request domain logic, constants, business-rule validation, balance calculations, and workflow state transitions.
- Reworked request services to support transactional operations and approval, rejection, and cancellation flows; added selector/query helpers.
- Corrected duplicate model code and foreign-key references, and added an initial Django migration.

Integration and access control

- Registered the backend app and routes and connected the frontend module in the application registry.
- Updated module dependency contracts, permission registry entries, and the generated permission catalog.

Reference materials

Added ELM problem statement, functional requirements, use cases, business rules, traceability matrix, and workflow specifications.

Improvement areas supported by the changes

Area	Evidence in the repository
Missing functionality	New ELM UI pages, API surfaces, domain logic, and migration.
Functional defects and business rules	Model corrections and explicit validation/rules for leave requests.
Workflow and state transitions	State machine and request approval, rejection, and cancellation flows.
Frontend/backend integration	Frontend API module, routes, types, and application registration.
Data model and configuration	Foreign-key corrections, initial migration, app and URL registration.
Validation and access control	Serializer/domain validation and permission registry/catalog updates.
Refactoring and duplication	Duplicate model code removed; services, selectors, and domain organized by responsibility.

Testing and verification caveat

The Phase-2 commit adds a test package initializer, but the recorded commit does not show new or updated test cases. This report therefore does not claim that additional tests were written or that a test suite passed.

Use of Cline / AI tools

Git history describes the earlier Phase-1 scaffold as generated via a “local code LLM.” The Phase-2 commit does not name Cline or record prompt logs. The repository can document the code changes, but cannot independently verify which AI tool was used, the division of AI and manual coding, the number of prompts, time sp

ent, or the exact original prompts. Those personal-study details should be suppl
ied from the author's own records rather than inferred.
Prepared from the project repository's commit history and change summary.